

FORM TP 2006159



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MAY/JUNE 2006

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ACCOUNTING

UNIT 2 – PAPER 02

2 $\frac{3}{4}$ hours

06 JUNE 2006 (p.m.)

1. This paper consists of THREE questions.
2. Each question is worth 35 marks.
3. ALL questions are compulsory.
4. Begin EACH answer on a new page.
5. Silent electronic calculators may be used, but ALL necessary working should be clearly shown.
6. **Present-value tables are provided in the appendix.**

1. (a) Ato Company is the leading manufacturer of premium quality tables in Guava Wood Country. The following data were extracted from the accounting records of Ato Company for the current period during which Ato produced 800 tables.

\$

Work-in-process inventory, beginning of year	32 400
Raw materials inventory, beginning of year	42 000
Finished goods inventory, beginning of year	90 000
Sales	1 376 400
* Indirect labour	80 400
Sales commissions expense	165 168
Raw materials purchased (net)	441 600
Rent – plant	20 640
Advertising expense	30 000
* Factory supplies used	41 400
Depreciation – office equipment (straight line)	59 400
Direct labour	236 400
* Plant utilities – electricity and water rates	16 440
Administrative office salaries	188 640
Insurance – plant	10 920
Depreciation – plant (units of output)	19 320
Work-in-process inventory, end of year	37 920
Raw materials inventory, end of year	46 800
Finished goods inventory, end of year	81 600

* Vary with units produced.

Compute the following for the current period:

- (i) Total variable costs incurred [4 marks]
- (ii) Total fixed costs incurred [3 marks]
- (iii) Total product costs incurred [4 marks]
- (iv) Direct materials used [3 marks]
- (v) Total manufacturing overhead [3 marks]
- (vi) Cost of goods manufactured [5 marks]
- (vii) Cost of goods sold [4 marks]

Note that a cost item may be used to compute more than one total.

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- (b) Ato company recently received an order from a customer in a neighbouring country, Mahogany Land, for 200 tables. The customer indicated that its regular supplier had experienced some minor industrial action and would be unable to meet its full requirement for the coming month. The customer further stated that the labour situation at its supplier had been resolved so this would be a one-time order.

The customer offers to pay Ato Company a flat fee of \$1 200 per dinette set and will pick up the merchandise at Ato Company's factory. If this one-time order is accepted, Ato Company will not pay commissions to any salesperson. Ato Company usually sells a table for \$1 800.

- (i) Assuming that Ato Company's fixed costs are constant within the range of 800 to 1300 dinette sets, should the order be accepted? **Show workings to support your answer.** [6 marks]
- (ii) List **THREE qualitative** factors that should be considered by a company when evaluating whether or not to accept a special order. [3 marks]

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2. Kingstown Industries manufactures two types of computerized scoreboards, custom and standard, each of which involves three manufacturing processes, namely, cutting, assembling, and packaging. The company uses an activity-based costing system to allocate its overhead cost to the manufacturing processes.

In using the activity-based costing system the company uses four overhead activities, production setup, procurement, quality control and materials management, in each production department.

An activity analysis of the overhead related to the cutting department revealed the following estimated costs and activity bases for the four overhead activities:

Activity	Cost	Activity Base
	\$	
Production setup	43 200	Number of setups
Procurement	156 000	Number of purchase orders
Quality control	175 000	Number of inspections
Materials management	<u>150 000</u>	Number of components
Total	<u>524 200</u>	

The activity-based usage quantities for **each** product in the cutting department are as follows:

	Setups	Purchase Orders	Inspections	Components	Unit Volume
Custom	400	1 000	2 000	300	5 000
Standard	80	200	500	100	5 000
Total	480	1 200	2 500	400	10 000

- Determine an activity rate for each activity. [4 marks]
- Calculate the total overhead cost assigned to each product in the cutting department using the rates computed in part (i). [8 marks]
- Determine the unit activity (overhead) cost for each product in the cutting department. [2 marks]

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The cutting department at Kingstown Industries uses a process-costing system to record the cost of processing its products. Currently, only the custom computerized scoreboard is made in this department. In arriving at the **conversion cost the company adds the overhead cost calculated at (ii) above, to the direct labour cost** incurred in the department.

The cutting department provided the following information relating to the custom computerized scoreboard for the current period.

Work-in-process, start of period, 1500 units, 35 per cent completed

	\$	\$
Direct materials	76 500	
Direct labour	12 400	
Factory overhead	42 700	131 600
Costs incurred in the current period		
Direct materials used		270 000
Direct labour cost		120 000
Factory overhead (determined from (ii) above)		
Units started in the current period		4 800
Ending work-in-process, 30 per cent completed		1 000

The cutting department uses the weighted average process-costing method and all direct materials are added at the beginning of production, while conversion activity occurs uniformly throughout the production process.

- (iv) Prepare a process-costing worksheet for the cutting department of Kingstown Industries for the current period using the weighted-average process costing method. **[21 marks]**

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3. (a) Red Gabby, owner of RG Wholesalers, is negotiating with the Caribbean Dominion Bank for a \$200 000, 12 per cent, 90-day loan, effective July 1 of the current year. If the bank grants the loan, the proceeds will be \$194 000, which Red Gabby intends to use on July 1 as follows: Pay accounts payable, \$150 000; purchase equipment, \$16 000; add to cash balance, \$28 000.

The current working capital position of RG Wholesalers, according to financial statements as of June 30, is as follows:

Cash in bank	\$ 20 000
Accounts receivable (net of allowance for doubtful accounts)	160 000
Merchandise inventory	90 000
Total current assets	\$270 000
Accounts payable (including accrued operating expenses)	150 000
Working capital	<u>\$120 000</u>

The bank loan officer asks Red Gabby to prepare a forecast of his cash receipts and cash payments for the next three months to demonstrate that the loan can be repaid at the end of September.

Red Gabby has made the following estimates which are to be used in preparing a three-month cash budget: Sales (all on credit) for July, \$300 000; August, \$360 000; September, \$270 000; October, \$200 000. Past experience indicates that 80 per cent of the receivables generated in any month will be collected in the month following the sale, 19 per cent in the second month following the sale, and 1 per cent will prove uncollectible. Red Gabby expects to collect \$120 000 of the June 30 receivables in July and the remaining \$40 000 in August.

Cost of goods sold has consistently averaged about 65 per cent of sales. Operating expenses are budgeted at \$36 000 per month plus 8 per cent of sales. With the exception of \$4 400 per month depreciation expense, all operating expenses and purchases are paid in the month following their incurrence.

Merchandise inventory at the end of **each** month should be sufficient to cover the following month's sales.

Prepare the following schedules for **each** of the three months, July, August and September

- (i) Cash collections on accounts receivable. [3 marks]
- (ii) Estimate merchandise purchased. [5 marks]
- (iii) Estimated cash payments for accounts payable (merchandise purchases) and operating expenses. [4 marks]
- (iv) Prepare a cash budget for **each** of the three months. [9 marks]
- (v) On the basis of the cash budget discuss whether it is likely that RG Wholesalers will be able to repay the \$200 000 loan to Caribbean Dominion Bank at the end of September. [1 mark]

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- (b) RG Wholesalers is also evaluating the acquisition of one of three possible pieces of equipment to automate its operations. The initial investment required for each of the proposed pieces of equipment is \$70 000 and they each have a useful life of 7 years. The firm's cost of capital is 14 per cent. The net cash flows for the three pieces of equipment are as follows:

Year	Equipment A	Equipment B	Equipment C
1	\$ 27 250	\$ 10 000	\$ 25 000
2	33 750	20 000	25 000
3	9 000	30 000	20 000
4	8 000	10 000	15 000
5	7 000	(8 000)	10 000
6	6 000	20 000	5 000
7	5 000	30 000	2 500
Total	\$ 96 000	\$112 000	\$102 500

- (i) To evaluate the investment opportunities compute the Net Present Value of the three pieces of equipment. **[12 marks]**
- (ii) Based on your calculations in part (i), which is the most attractive opportunity for RG Wholesalers?

Present Value Table to be used with question 3(b)

Year	12%	14%	16%
1	0.8929	0.8772	0.8621
2	0.7972	0.7695	0.7432
3	0.7118	0.6750	0.6407
4	0.6355	0.5921	0.5523
5	0.5674	0.5194	0.4761
6	0.5066	0.4556	0.4104
7	0.4523	0.3996	0.3538

[1 mark]

END OF TEST